A/B Testing Performance Report

Impala Version Comparison

Comparing Apache Impala 4.3.0 vs 4.5.0

Report Generated: 2026-04-03 16:31:59

Introduction

This report presents a comprehensive analysis of A/B testing results comparing the performance of two Apache Impala versions:

• **Sample A**: apache_impalad_version_4.3.0_performance.csv
• **Sample B**: apache_impalad_version_4.5.0_performance.csv

A/B testing is a statistical method used to compare two variants to determine which one performs better. In this context, we're measuring query execution performance to identify if apache_impalad_version_4.5.0_performance.csv offers performance differences compared to apache_impalad_version_4.3.0_performance.csv.

Methodology

Statistical Test Used: Two-sample paired t-test
Purpose: Compare means of two related groups (paired samples from the same queries)

Hypotheses:
• **Null Hypothesis (H■):** There is no significant difference in query execution times between Impala 4.3.0 and 4.5.0
• **Alternative Hypothesis (H■):** There is a significant difference in query execution times

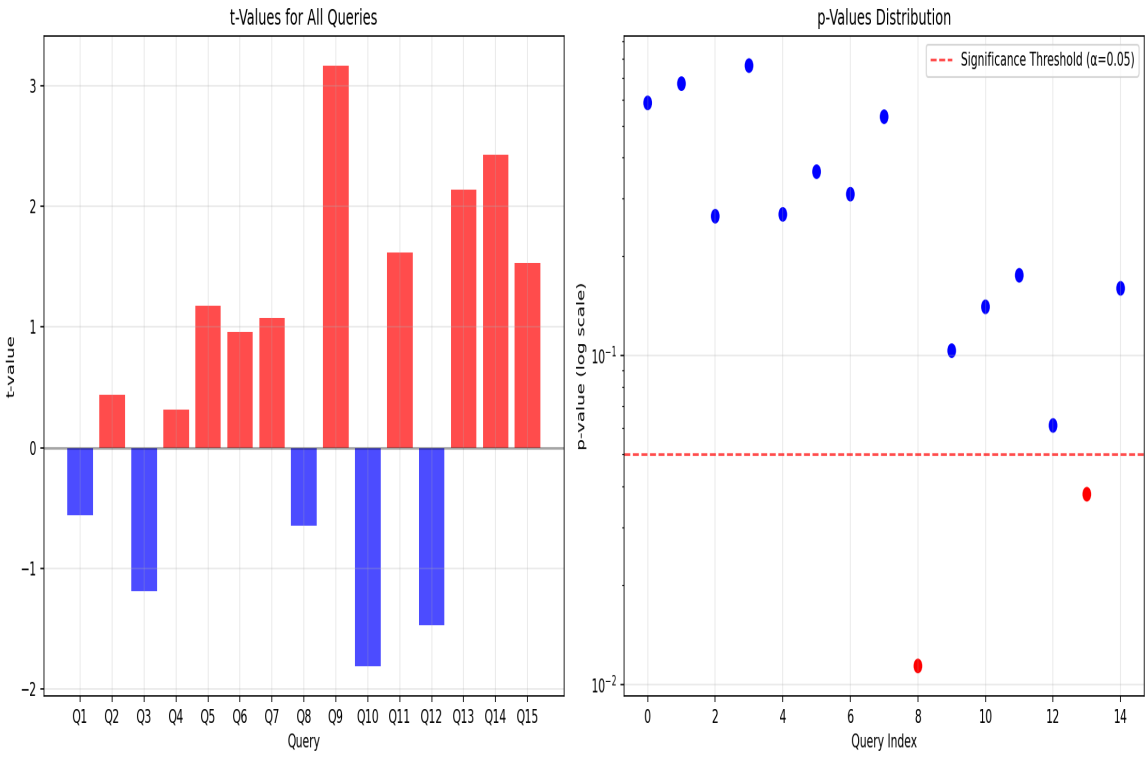
Significance Level: α (alpha) = 0.05 (5% significance level)
• If p-value < 0.05, we reject H■ and conclude there's a significant difference
• If p-value ≥ 0.05, we fail to reject H■ (no significant difference detected)

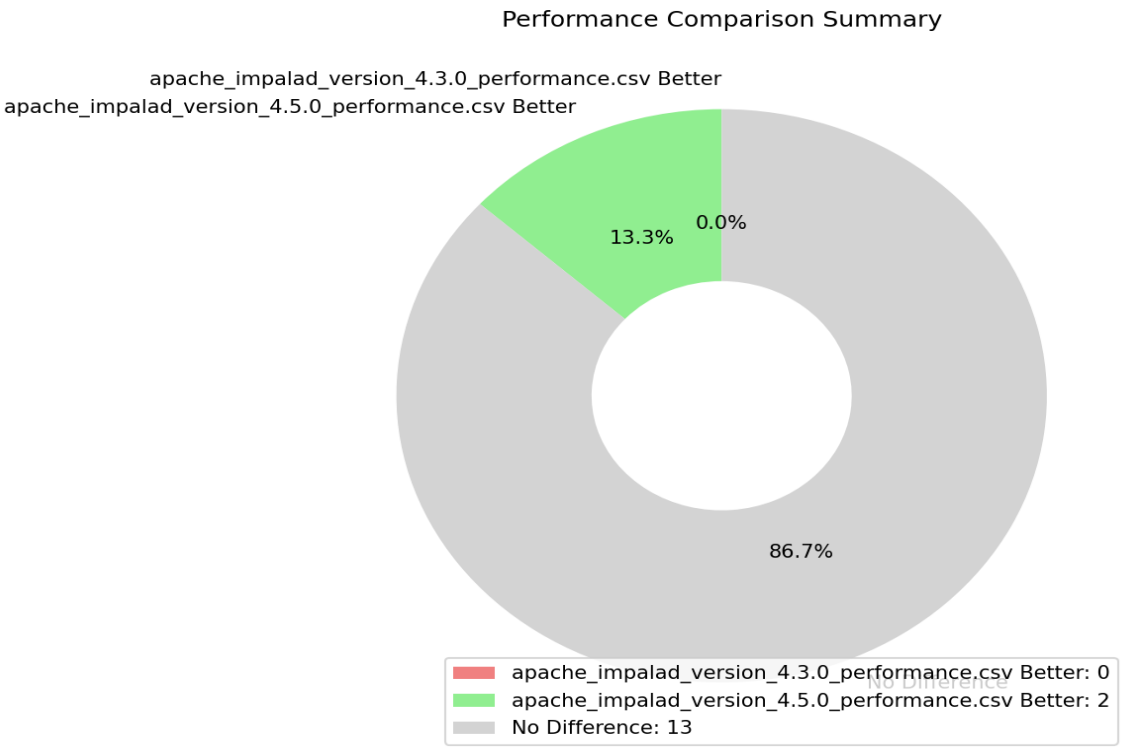
Interpretation of Results:
• **t-value:** Measures the difference between group means relative to the variation within groups
• Positive t-value: Sample A (4.3.0) is slower than Sample B (4.5.0)
• Negative t-value: Sample A (4.3.0) is faster than Sample B (4.5.0)
• **p-value:** Probability of observing the data assuming H■ is true

Summary Statistics

Metric	Value
Total queries tested	15
Queries with significant performance difference	2 (13.3%)
Queries with no significant difference	13 (86.7%)
Queries where 4.3.0 performs better	0
Queries where 4.5.0 performs better	2

Overall Performance Analysis

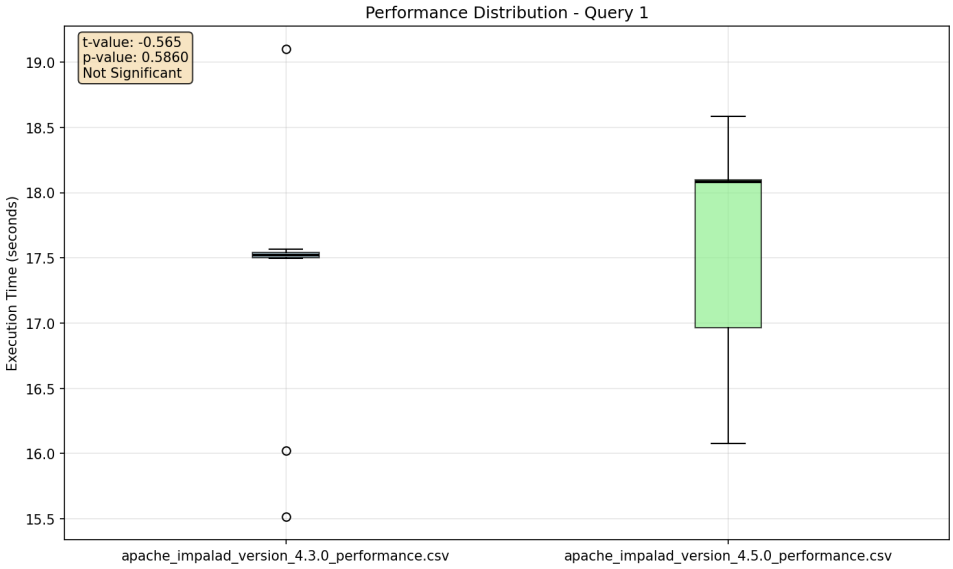




Detailed Results by Query

Query 1

Query: MT_DOP=0;
SELECT
status, COUNT(*) AS tx_count
FROM
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
GROUP BY
status;



Statistical Results:

Metric	Value
t-value	-0.564822
p-value	0.585995
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	17.331887	17.637890
median	17.520179	18.083035
std	0.965631	0.980014
min	15.517450	16.076462

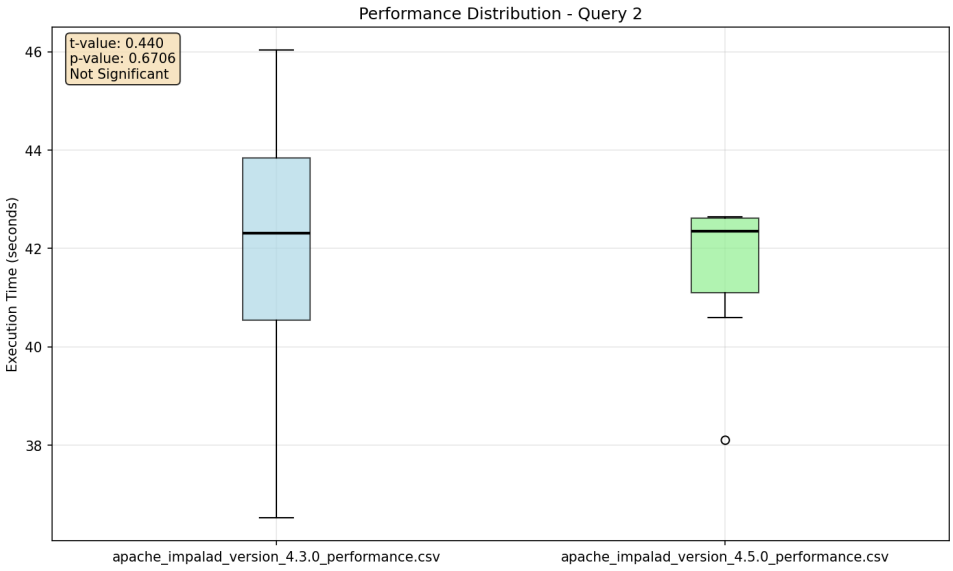
max	19.098637	18.583935

Detailed Analysis:

No significant difference (p-value = 0.585995 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 2

Query: MT_DOP=0;
SELECT
op_code, bound_type, status,
COUNT(*) AS total_tx,
COUNT(DISTINCT msisdn) AS unique_subs,
COUNT(DISTINCT mcc) AS unique_visited_mcc
FROM
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
AND par_date BETWEEN 20240601 AND 20240610
GROUP BY
op_code, bound_type, status
ORDER BY
total_tx DESC;



Statistical Results:

Metric	Value
t-value	0.439535

p-value	0.670638
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	42.053146	41.607775
median	42.306942	42.351241
std	2.752786	1.456698
min	36.524794	38.107227
max	46.029868	42.644414

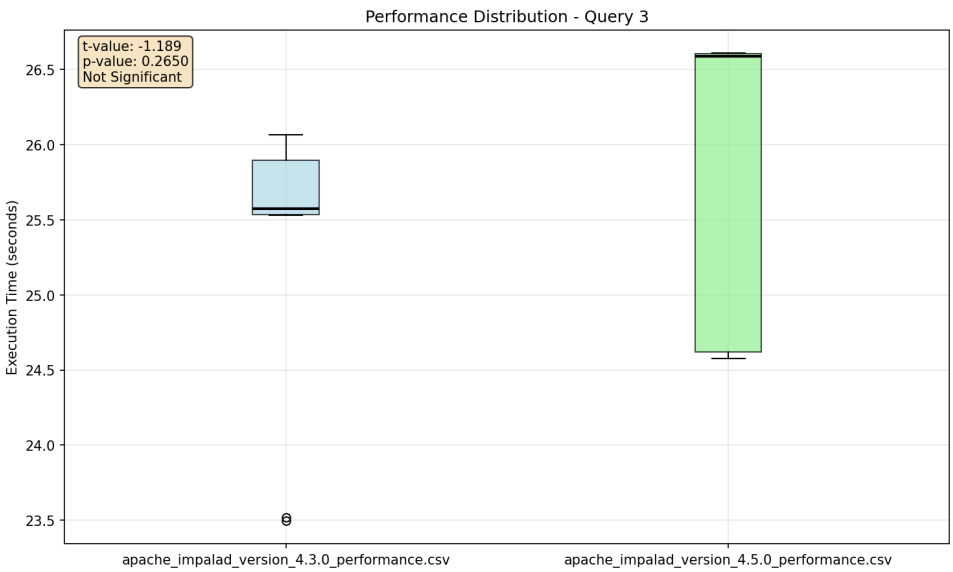
Detailed Analysis:

No significant difference (p-value = 0.670638 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 3

Query: MT_DOP=0;

```
SELECT
tx_month, tx_week, op_code, status,
COUNT(*) AS tx_count,
SUM(CASE WHEN mcc != mcc_ref THEN 1 ELSE 0 END) AS roaming_cnt
FROM
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
GROUP BY
tx_month, tx_week, op_code, status
ORDER BY
tx_month, tx_week;
```



Statistical Results:

Metric	Value
t-value	-1.188679
p-value	0.264983
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

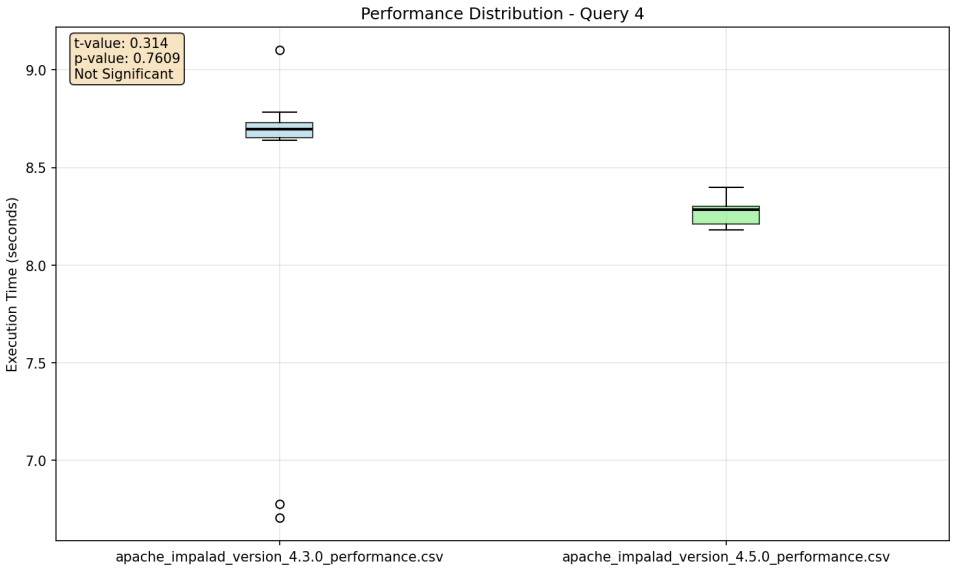
	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	25.293790	25.800199
median	25.572621	26.587177
std	0.964875	1.032373
min	23.498270	24.577496
max	26.064650	26.608793

Detailed Analysis:

No significant difference (p-value = 0.264983 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 4

Query: MT_DOP=0;
SELECT tx_hour, op_code, COUNT(*) AS tx_count
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND tx_hour BETWEEN 8 AND 10 -- narrow: ~12.5% of hours
GROUP BY tx_hour, op_code
ORDER BY tx_hour;



Statistical Results:

Metric	Value
t-value	0.313717
p-value	0.760886
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

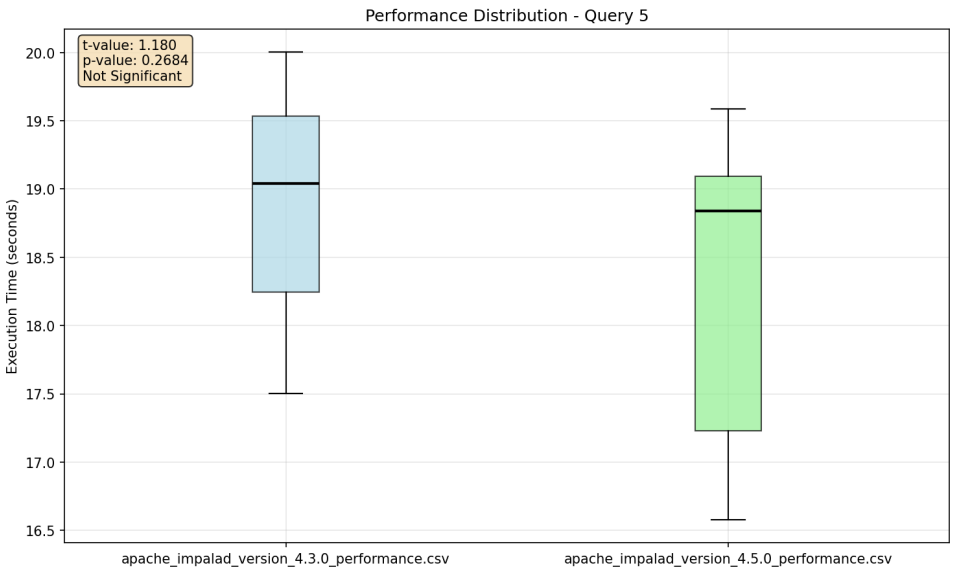
	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	8.353648	8.270236
median	8.695215	8.285501
std	0.858779	0.068264
min	6.707406	8.178964
max	9.099857	8.396842

Detailed Analysis:

No significant difference (p-value = 0.760886 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 5

Query: MT_DOP=0;
SELECT a.msisdn,
a.mcc,
a.op_code,
COUNT(*) AS tx_count
FROM roam352_report_digi.data_em a
JOIN (
SELECT DISTINCT msisdn
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND module_dt BETWEEN 1717545900000 AND 1717545960000 -- 1-minute window
) narrow ON a.msisdn = narrow.msisdn
WHERE a.par_year = 2024
AND a.par_month = 202406
GROUP BY a.msisdn, a.mcc, a.op_code
ORDER BY tx_count DESC
LIMIT 50;



Statistical Results:

Metric	Value
t-value	1.179564
p-value	0.268413
Reject H■	No

Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	18.875807	18.296124
median	19.042719	18.841528
std	0.885247	1.201108
min	17.501208	16.579278
max	20.001613	19.588318

Detailed Analysis:

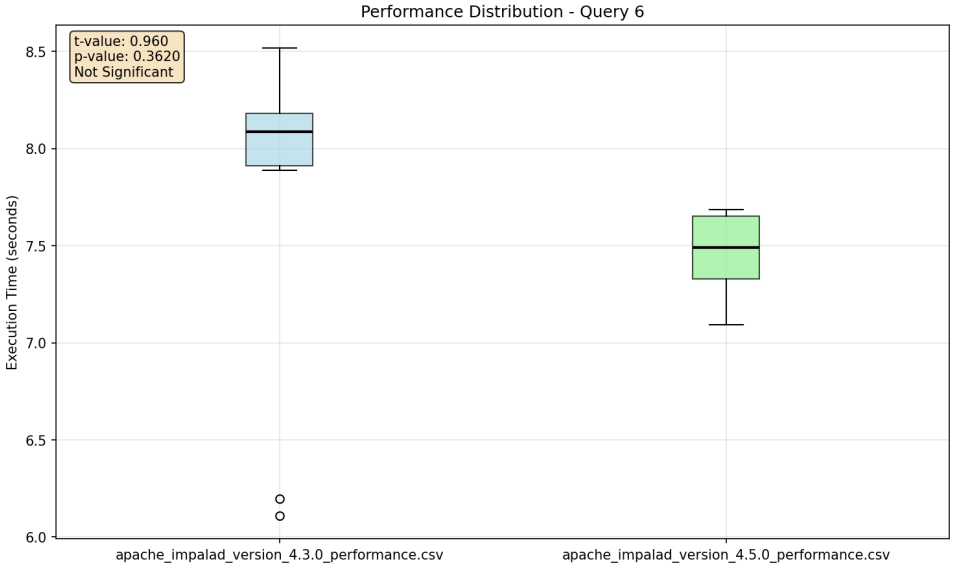
No significant difference (p-value = 0.268413 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 6

Query: MT_DOP=0;

```
SELECT mcc, rank_in_mcc, COUNT(*) AS sub_count
FROM (
SELECT msisdn, mcc, mnc, pct_roaming, rank_in_mcc
FROM (
SELECT msisdn, mcc, mnc, pct_roaming,
RANK() OVER (PARTITION BY mcc ORDER BY pct_roaming DESC) AS rank_in_mcc
FROM (
SELECT msisdn, mcc, mnc,
roaming_cnt * 100.0 / NULLIF(total_tx, 0) AS pct_roaming
FROM (
SELECT msisdn, mcc, mnc,
total_tx,
roaming_cnt,
total_tx - roaming_cnt AS non_roaming_cnt
FROM (
SELECT msisdn, mcc, mnc,
COUNT(*) AS total_tx,
SUM(CASE WHEN mcc != mcc_ref THEN 1 ELSE 0 END) AS roaming_cnt
FROM (
SELECT msisdn, mcc, mnc, mcc_ref, op_code, status
FROM (
SELECT msisdn, mcc, mnc, mcc_ref,
op_code, status, bound_type
FROM (
SELECT msisdn, mcc, mnc, mcc_ref,
op_code, status, bound_type, vlr
```

```
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND par_date = 20240601
AND msisdn != 0
) s1
WHERE bound_type IN (0, 1)
) s2
WHERE op_code IN (2, 3)
) s3
GROUP BY msisdn, mcc, mnc, mcc_ref
) s4
) s5
) s6
) s7
WHERE rank_in_mcc <= 10
) s8
GROUP BY mcc, rank_in_mcc
ORDER BY mcc, rank_in_mcc;
```



Statistical Results:

Metric	Value
t-value	0.960384
p-value	0.361951
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv

mean	7.737899	7.463382
median	8.087561	7.490949
std	0.851748	0.214370
min	6.110288	7.094304
max	8.515823	7.685838

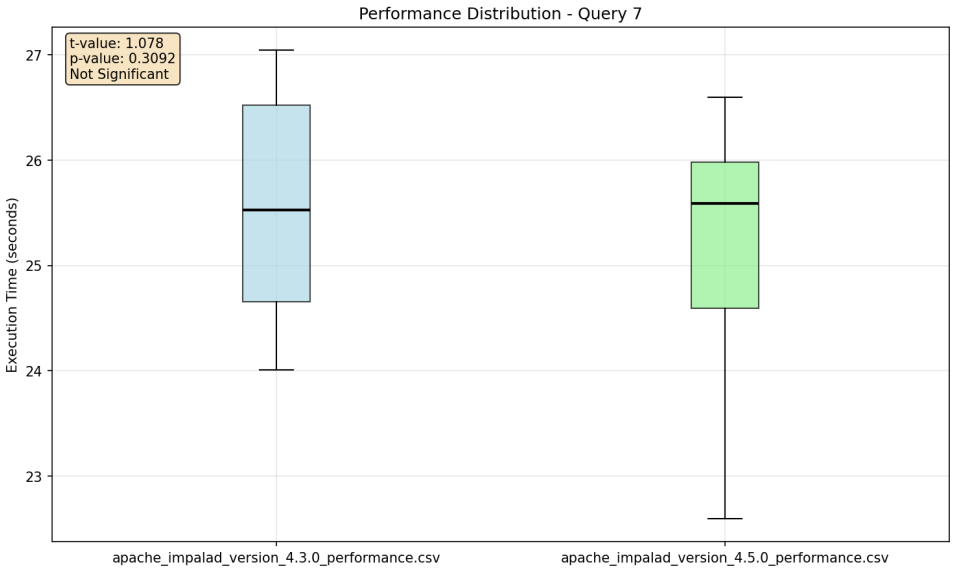
Detailed Analysis:

No significant difference (p-value = 0.361951 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 7

Query: MT_DOP=0;

```
SELECT final.*
FROM (
SELECT msisdn, tx_hour, tx_count, running_total,
ROUND(running_total * 100.0 / NULLIF(day_total, 0), 2) AS cumulative_pct
FROM (
SELECT msisdn, tx_hour, tx_count,
SUM(tx_count) OVER (PARTITION BY msisdn ORDER BY tx_hour) AS running_total,
SUM(tx_count) OVER (PARTITION BY msisdn) AS day_total
FROM (
SELECT msisdn, tx_hour, COUNT(*) AS tx_count
FROM (
SELECT msisdn, tx_hour, op_code, status
FROM (
SELECT *
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND par_date BETWEEN 20240601 AND 20240615
) base
WHERE op_code IN (2, 3)
) filtered
GROUP BY msisdn, tx_hour
) hourly
) windowed
) final
WHERE msisdn != 0
ORDER BY msisdn, tx_hour
LIMIT 200;
```



Statistical Results:

Metric	Value
t-value	1.077738
p-value	0.309188
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

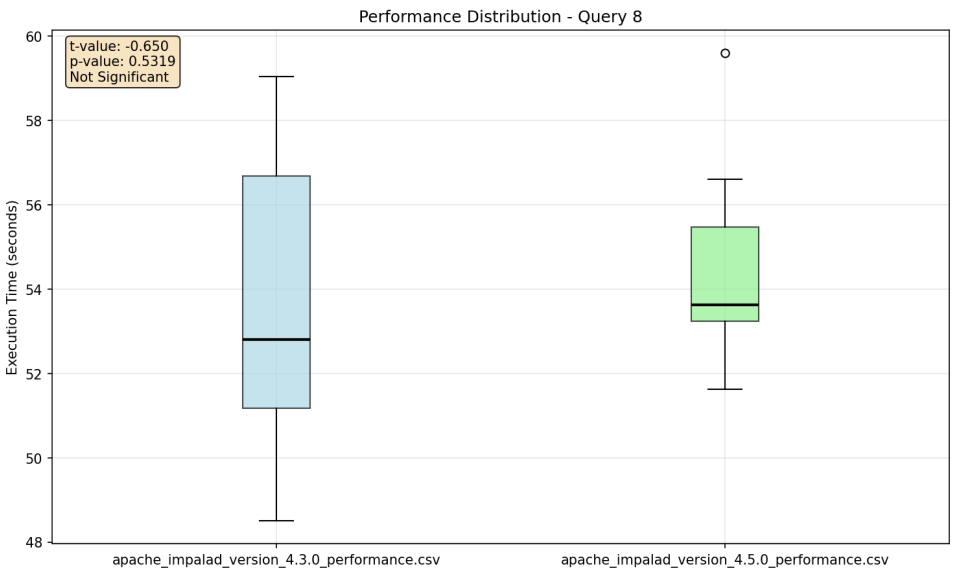
	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	25.619333	25.149495
median	25.528718	25.589851
std	1.102289	1.189534
min	24.011250	22.599058
max	27.040897	26.593363

Detailed Analysis:

No significant difference (p-value = 0.309188 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 8

Query: MT_DOP=0;
SELECT
mcc, op_code, COUNT(*) AS tx_count, COUNT(DISTINCT msisdn) AS unique_subs
FROM
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
GROUP BY
mcc, op_code;



Statistical Results:

Metric	Value
t-value	-0.649998
p-value	0.531932
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	53.696641	54.555948
median	52.801898	53.621072
std	3.445840	2.273697
min	48.511176	51.618730
max	59.048396	59.593353

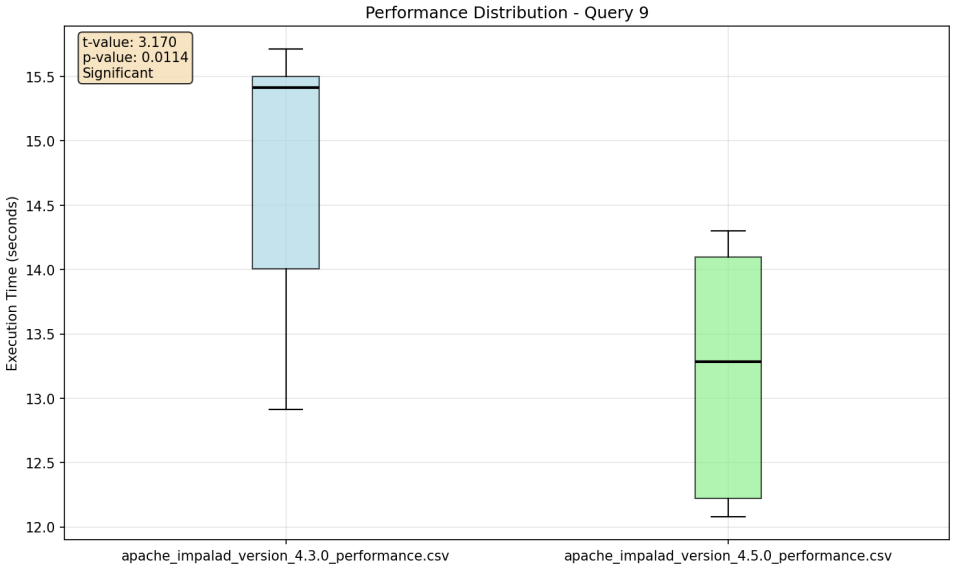
Detailed Analysis:

No significant difference (p-value = 0.531932 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 9

Query: MT_DOP=0;
SELECT
mcc, op_code, SUM(CASE WHEN status = 'SUCCESS' THEN 1 ELSE 0 END) AS success_cnt
FROM

```
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
AND par_date BETWEEN 20240601 AND 20240615
GROUP BY
mcc, op_code;
```



Statistical Results:

Metric	Value
t-value	3.170297
p-value	0.011361
Reject H■	Yes
Conclusion	Sample A is slower than Sample B

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	14.846947	13.191551
median	15.416059	13.283905
std	1.065761	1.021101
min	12.912141	12.080717
max	15.711941	14.299141

Detailed Analysis:

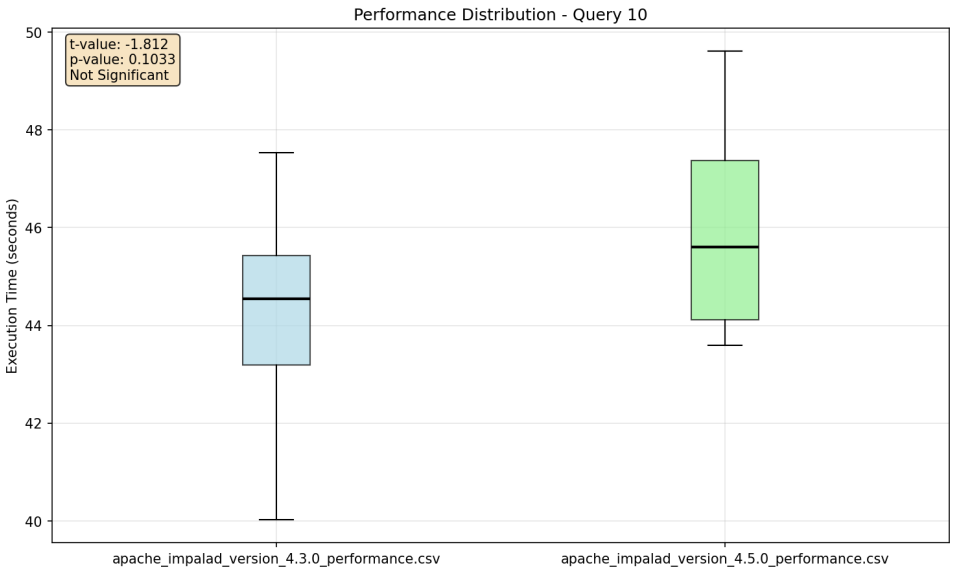
Significant difference detected (p-value = 0.011361 < 0.05)

Performance comparison: Impala 4.3.0 executes this query significantly slower than Impala 4.5.0

Implication: Version 4.5.0 shows performance improvement for this query

Query 10

Query: MT_DOP=0;
SELECT mcc, bound_type,
COUNT(*) AS total_tx,
COUNT(DISTINCT imsi) AS unique_imsi,
COUNT(DISTINCT vlr) AS unique_vlr
FROM
roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
AND par_date BETWEEN 20240601 AND 20240615
GROUP BY
mcc, bound_type
ORDER BY
total_tx DESC;



Statistical Results:

Metric	Value
t-value	-1.812367
p-value	0.103346
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	44.243132	45.963008
median	44.546905	45.603334
std	2.088470	2.114959

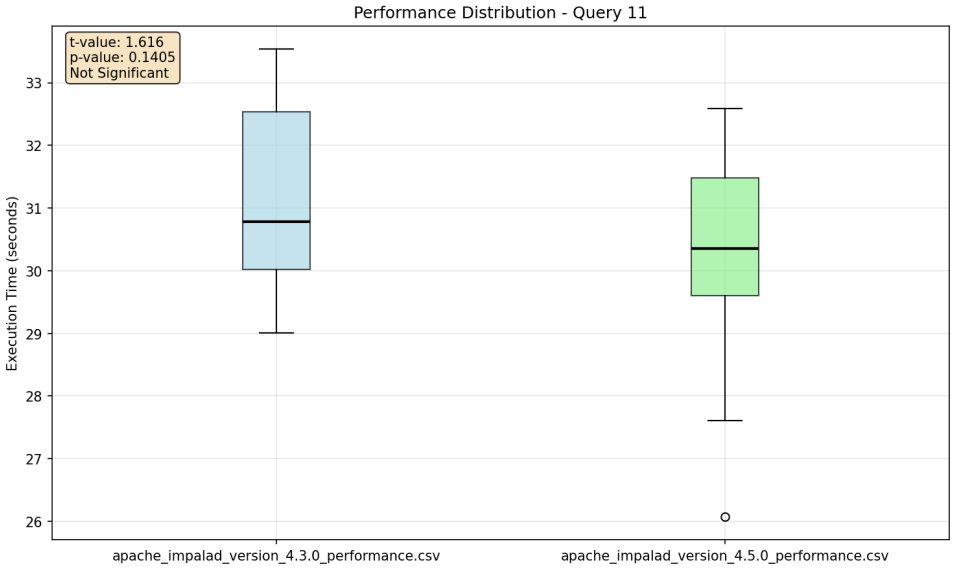
min	40.028371	43.596459
max	47.540738	49.608878

Detailed Analysis:

No significant difference (p-value = 0.103346 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 11

Query: MT_DOP=0;
SELECT op_code,
tx_hour,
COUNT(*) AS total_tx,
COUNT(DISTINCT msisdn) AS unique_subs
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND op_code = 2 -- single value predicate → agg cardinality reduced
GROUP BY op_code, tx_hour
ORDER BY tx_hour;



Statistical Results:

Metric	Value
t-value	1.616262
p-value	0.140493
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

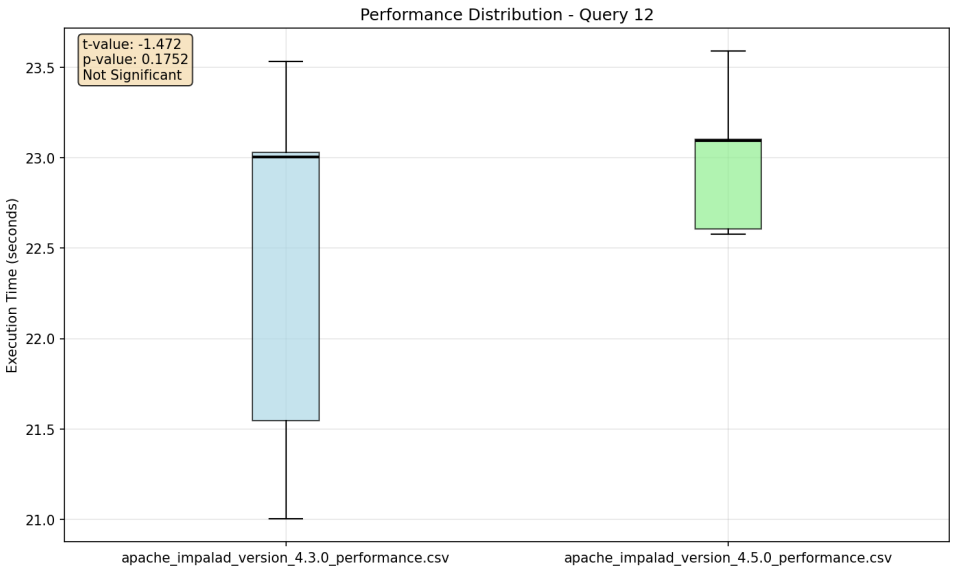
	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	31.186853	30.098560
median	30.783524	30.350367
std	1.597773	2.017119
min	29.007377	26.076402
max	33.535041	32.593094

Detailed Analysis:

No significant difference (p-value = 0.140493 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 12

Query: MT_DOP=0;
SELECT bound_type, status,
COUNT(*) AS tx_count,
COUNT(DISTINCT mcc) AS unique_mcc
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND bound_type IN (0, 1) -- subset of all bound_type values
GROUP BY bound_type, status;



Statistical Results:

Metric	Value

t-value	-1.471618
p-value	0.175206
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

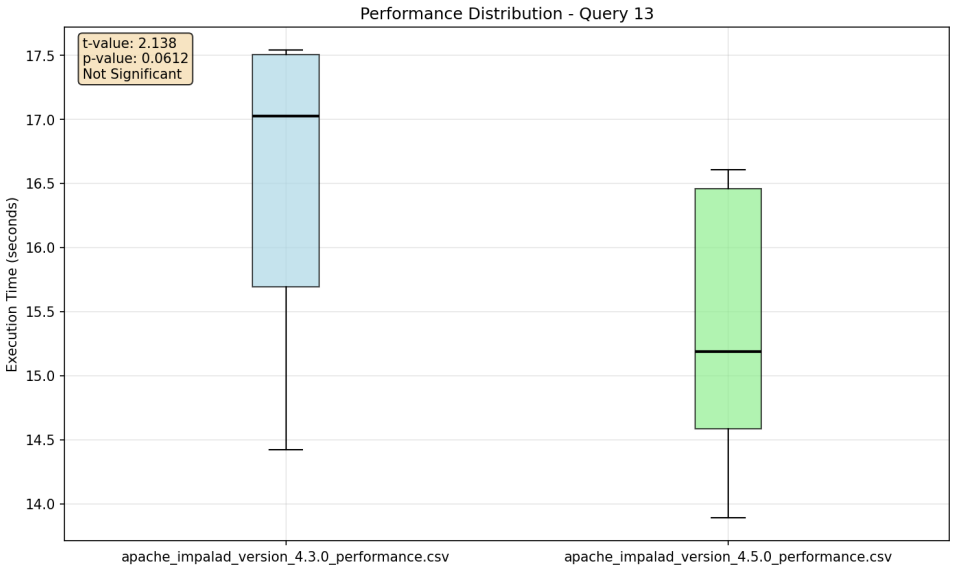
	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	22.470962	22.946583
median	23.005076	23.092858
std	1.010361	0.337805
min	21.004827	22.578605
max	23.533618	23.588654

Detailed Analysis:

No significant difference (p-value = 0.175206 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 13

Query: MT_DOP=0;
SELECT visited.mcc,
home.mcc_ref,
COUNT(*) AS tx_count,
COUNT(DISTINCT visited.msisdn) AS unique_subs
FROM roam352_report_digi.data_em visited
JOIN (
SELECT DISTINCT msisdn, mcc_ref
FROM roam352_report_digi.data_em
WHERE par_year = 2024
AND par_month = 202406
AND mcc_ref = 502
) home ON visited.msisdn = home.msisdn
WHERE visited.par_year = 2024
AND visited.par_month = 202406
GROUP BY visited.mcc, home.mcc_ref;



Statistical Results:

Metric	Value
t-value	2.138083
p-value	0.061212
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	16.476330	15.390807
median	17.025015	15.187550

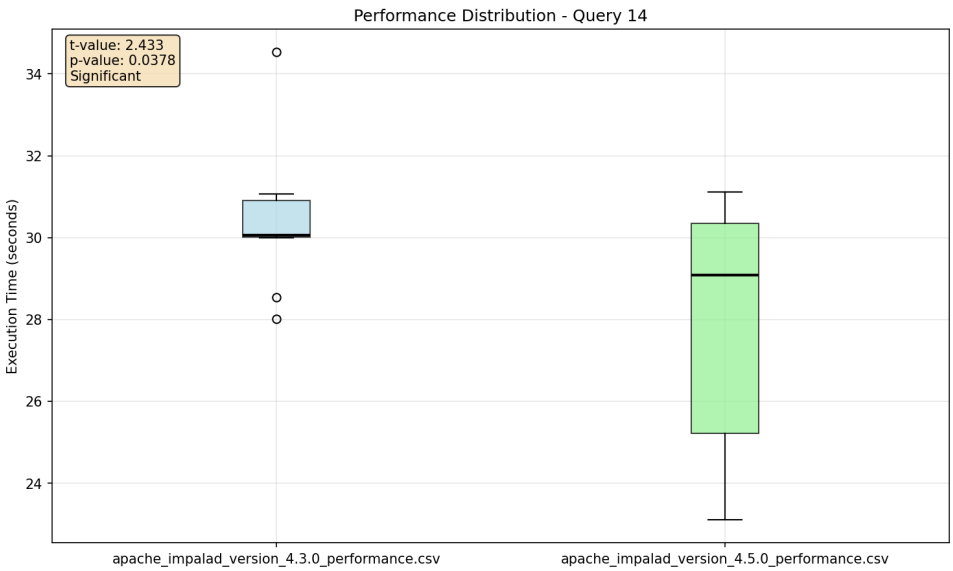
std	1.248469	1.044477
min	14.420981	13.892244
max	17.537340	16.605692

Detailed Analysis:

No significant difference (p-value = 0.061212 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Query 14

Query: MT_DOP=4;
SELECT msisdn,
imsi,
COUNT(*) AS total_tx,
COUNT(DISTINCT vlr) AS vlr_hops,
COUNT(DISTINCT mcc) AS mcc_visited,
SUM(CASE WHEN status = 'SUCCESS' THEN 1 ELSE 0 END) AS success_cnt,
SUM(CASE WHEN mcc != mcc_ref THEN 1 ELSE 0 END) AS roaming_tx
FROM roam352_report_digi.data_em
WHERE
par_year = 2024
AND par_month = 202406
AND par_date BETWEEN 20240601 AND 20240615
AND msisdn != 0
GROUP BY msisdn, imsi
ORDER BY total_tx DESC
LIMIT 500;



Statistical Results:

Metric	Value
t-value	2.432615
p-value	0.037820
Reject H■	Yes
Conclusion	Sample A is slower than Sample B

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	30.380892	27.947369
median	30.052003	29.086657
std	1.752848	3.111849
min	28.006426	23.105028
max	34.528342	31.106919

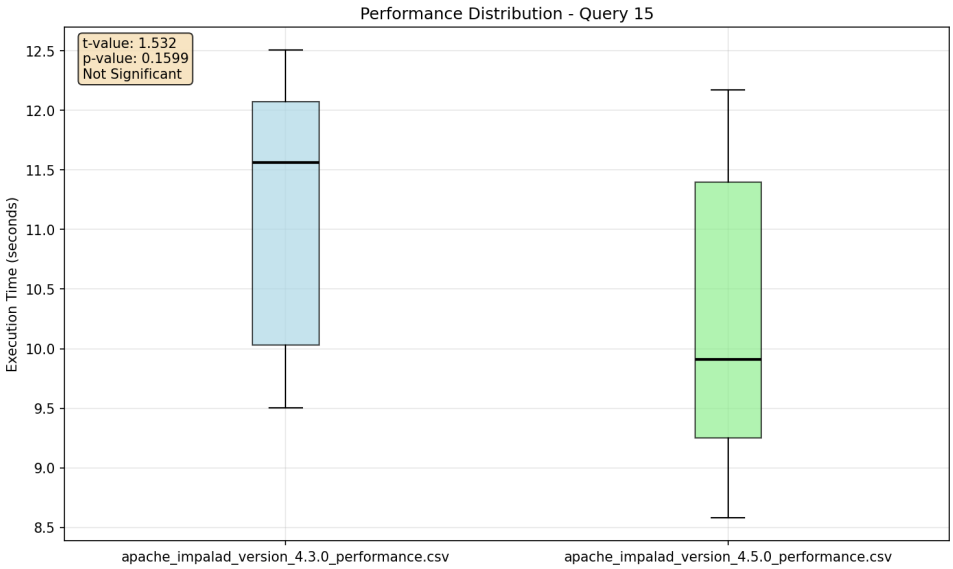
Detailed Analysis:

Significant difference detected (p-value = 0.037820 < 0.05)
Performance comparison: Impala 4.3.0 executes this query significantly slower than Impala 4.5.0
Implication: Version 4.5.0 shows performance improvement for this query

Query 15

Query: MT_DOP=4;
SELECT a.msisdn,
a.mcc,
a.vlr,
b.op_code AS paired_op_code,
COUNT(*) AS pair_count
FROM roam352_report_digi.data_em a
JOIN roam352_report_digi.data_em b
ON a.msisdn = b.msisdn
AND a.par_year = b.par_year
AND a.par_month = b.par_month
AND a.op_code != b.op_code
WHERE
a.par_year = 2024
AND a.par_month = 202406
AND a.par_date = 20240601
AND b.par_year = 2024
AND b.par_month = 202406
AND b.par_date = 20240601
AND a.op_code IN (23, 316)
AND b.op_code IN (23, 316)

```
AND a.mcc IN (502, 510)
AND b.mcc IN (502, 510)
AND a.msisdn != 0
AND b.msisdn != 0
GROUP BY a.msisdn, a.mcc, a.vlr, b.op_code
LIMIT 200;
```



Statistical Results:

Metric	Value
t-value	1.531930
p-value	0.159902
Reject H■	No
Conclusion	Both samples has no significant difference

Recorded performance (execution times) for this query:

	apache_impalad_version_4.3.0_performance.csv	apache_impalad_version_4.5.0_performance.csv
mean	11.133376	10.291613
median	11.560461	9.907140
std	1.158759	1.299424
min	9.501885	8.582248
max	12.504588	12.174048

Detailed Analysis:

No significant difference (p-value = 0.159902 ≥ 0.05)
Performance comparison: Both versions perform similarly for this query
Implication: No clear performance advantage for either version

Overall Conclusion

Significant performance differences were detected in 2 out of 15 queries (13.3%).**
Overall, **Impala 4.5.0 shows better performance** in the majority of queries where differences were significant.

Recommendation: Consider upgrading to Impala 4.5.0 for potential performance improvements, but test specific query patterns in your environment before deployment.